



XXVI INTERNATIONAL SYMPOSIUM ON SOLID STATE DOSIMETRY

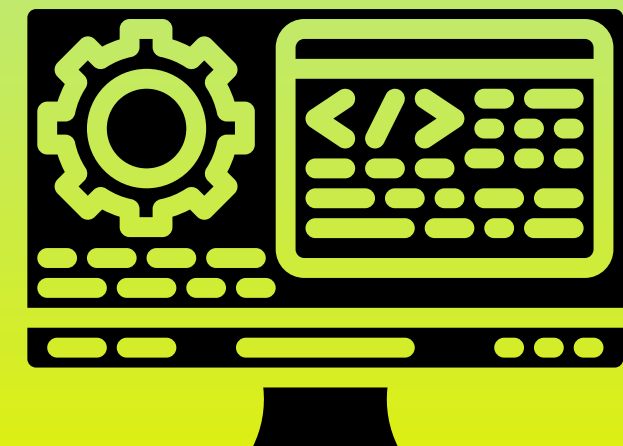


IMPACT OF CRYSTAL GEOMETRY SIMPLIFICATION ON HPGe DETECTOR RESPONSE: A MONTE CARLO STUDY

PRESENTED BY VÍCTOR BERNARDES DE MORAIS

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INTRODUCTION

Why Detector Calibration Matters



Full-energy peak efficiency is a fundamental parameter in gamma-ray spectrometry

Accurate quantification of radionuclide activity with HPGe detectors depends directly on precisely knowing the full-energy peak efficiency. It is essential for radiological protection, environmental monitoring, and nuclear engineering.



Experimental calibration is necessary and serves as the reference, but it has real limits

It requires certified point sources spanning the energy range of interest: high cost, limited availability, and restricted energy coverage, particularly when several geometries must be calibrated.



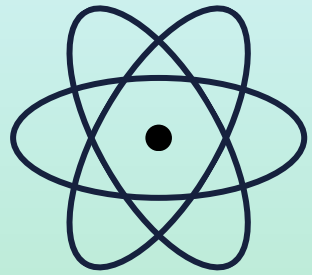
Monte Carlo simulation (MCNP6) offers a flexible alternative

It allows efficiency to be estimated for arbitrary source-detector geometries and energies, but its reliability depends strongly on how accurately the detector geometry is modeled.



INTRODUCTION

The Gap and The Objective



Rounded Crystal Edges: A Commonly Simplified Geometric Detail

Commercial HPGe crystals are manufactured with rounded edges, a real geometric feature that changes the effective active volume seen by incoming photons, particularly at low energies and near the crystal periphery.

Yet this detail is frequently replaced by simplified flat peripheral surfaces in MCNP models, because representing it explicitly considerably increases modeling complexity.

The impact of this common simplification on simulated full-energy peak efficiency has received limited systematic investigation.



Objective

Evaluate the influence of crystal edge geometry simplification on the simulated full-energy peak efficiency of a GC2520 HPGe detector, over the extended energy range provided by the multiple γ -ray emissions of ^{152}Eu .

Two MCNP6 models were developed, differing only in crystal periphery and electrode-bore geometry.



MATERIALS AND METHODS

Detector and Experimental Setup



HPGe Detector

Canberra GC2520 coaxial p-type detector. Crystal Ø 62.00 mm, length 34.60 mm. Relative efficiency $\geq 25\%$, resolution ≤ 2.0 keV FWHM at 1332 keV.



^{152}Eu Point Source

Calibrated activity of 60.277 kBq, positioned on the detector symmetry axis at 20.456 cm. Its multiple emission lines provide efficiency data from 0.12178 to 1.40795 MeV.



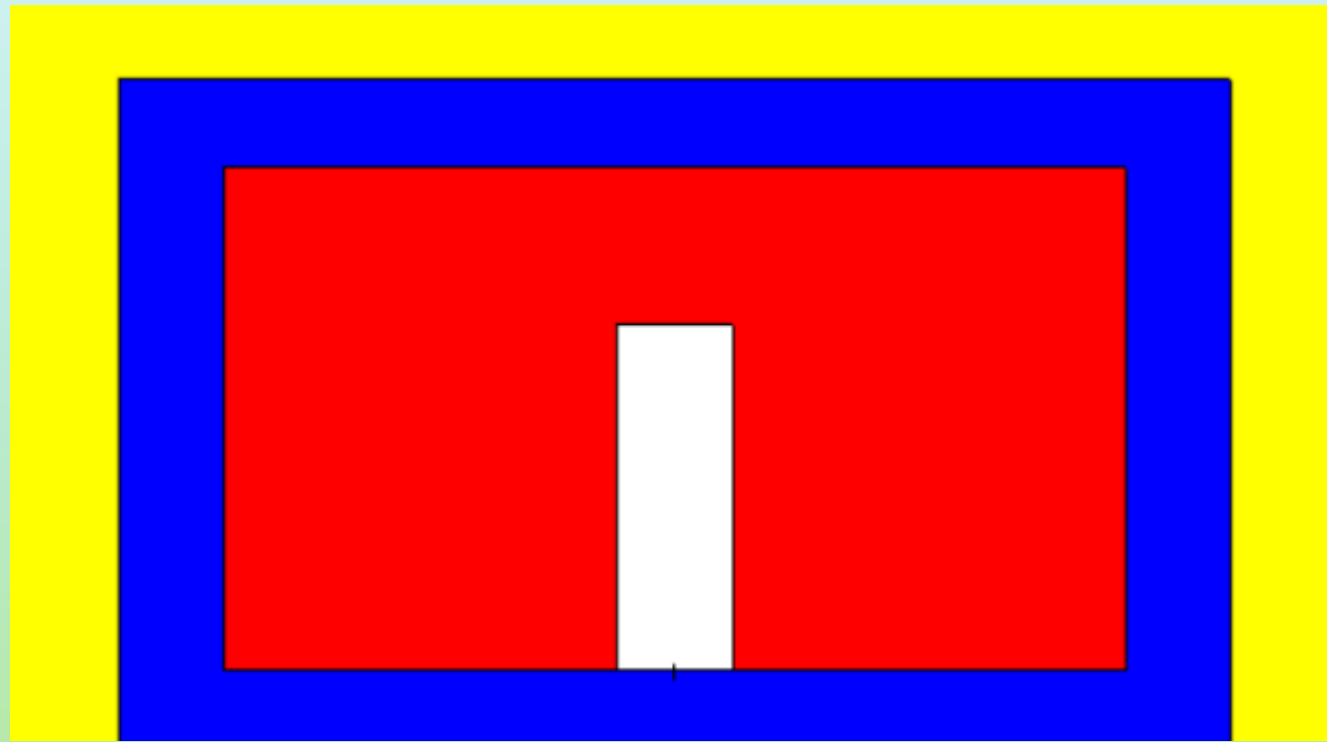
Data Acquisition

Genie 2000 software. 24-hour counting live time (dead time 0.84%). Efficiency obtained from net peak area, decay-corrected activity, and emission probability.

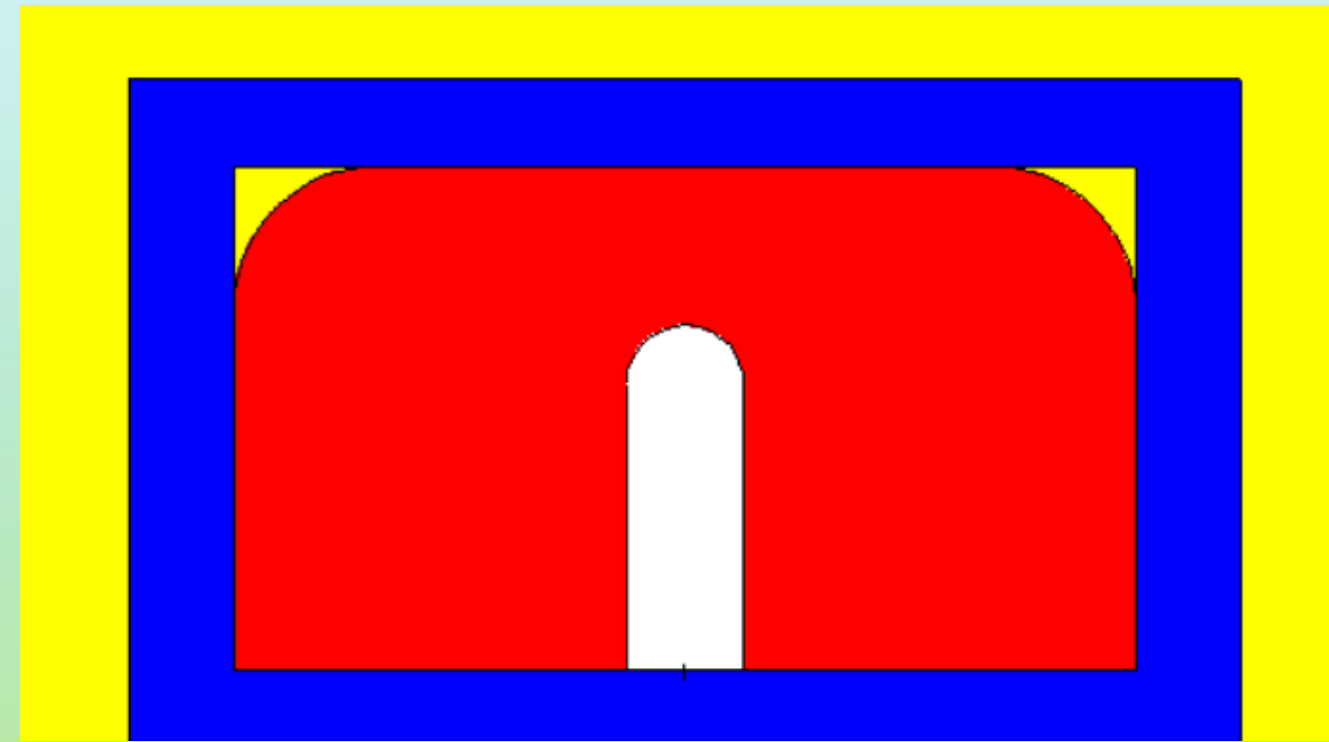


MATERIALS AND METHODS

Two MCNP6 Detector Models



(a) Basic model - flat edges



(b) Advanced model - rounded edges

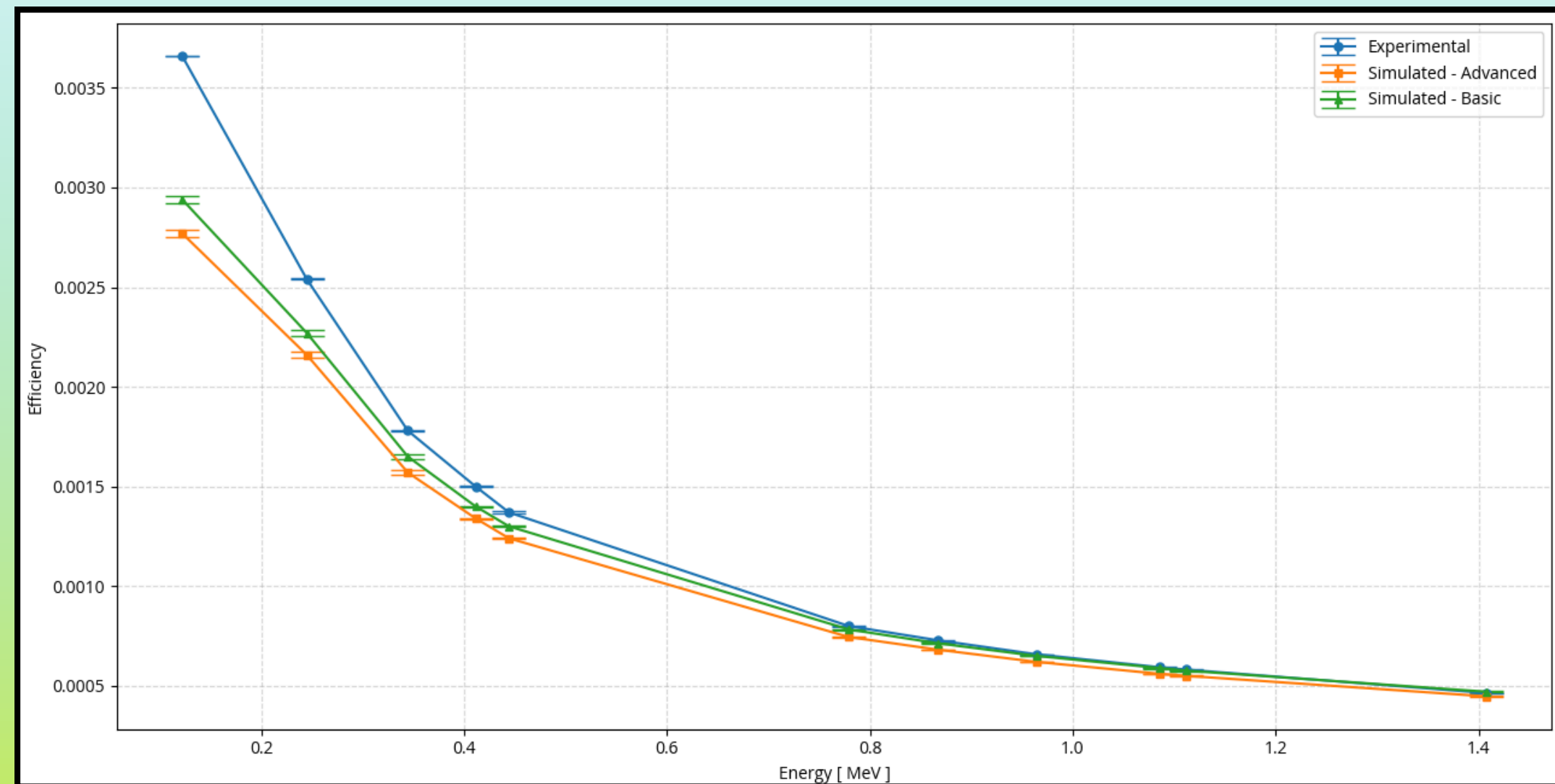
- Only the crystal periphery and the electrode-bore geometry differ between the two models, with all other parameters are identical.
- Electrode size was refined once, iteratively, against the advanced model at the highest-energy line (1.40795 MeV), then applied unchanged to both models.
- Neither model includes the detector's dead layer, isolating the comparison on the nominal manufacturer geometry.
- Photon and electron transport (MODE P E), pulse-height tally (F8), and 10^8 histories (NPS 1E8) for relative uncertainty $< 1\%$.



RESULTS

Full-Energy Peak Efficiency Curves

Both simulated curves reproduce the shape of the experimental curve: a steep decrease at low energies, then a gradual decline at higher energies.



Experimental and simulated (basic and advanced models) full-energy peak efficiency curves for ^{152}Eu .

The largest discrepancies between simulated and experimental occur at 0.12178 MeV: **19.54% for the basic model** and **24.45% for the advanced model**.



RESULTS

A Comparative of Simulated vs Experimental

Table 1.- Simulated (basic model) and experimental full-energy peak efficiencies for ¹⁵² Eu.			
Energy [MeV]	Experimental Efficiency (x10 ⁻³)	Simulated – Basic Efficiency (x10 ⁻³)	Discrepancy (%)
0.12178	3.660±0.002	2.940±0.017	19.54
0.24469	2.540±0.003	2.270±0.015	10.52
0.34427	1.780±0.001	1.650±0.013	7.08
0.41111	1.500±0.004	1.400±0.004	6.54
0.44398	1.370±0.004	1.300±0.004	4.91
0.77889	0.800±0.001	0.783±0.003	2.12
0.86732	0.728±0.002	0.713±0.003	2.11
0.96401	0.658±0.001	0.650± 0.003	1.17
1.08578	0.593±0.001	0.589±0.002	0.68
1.11202	0.581±0.001	0.576±0.002	0.78
1.40795	0.464±0.001	0.470±0.002	1.25

The Basic Model demonstrates strong agreement with experimental data at medium and high energies (discrepancy < 2% above 0.78 MeV), with an overall average discrepancy of 5.2%.

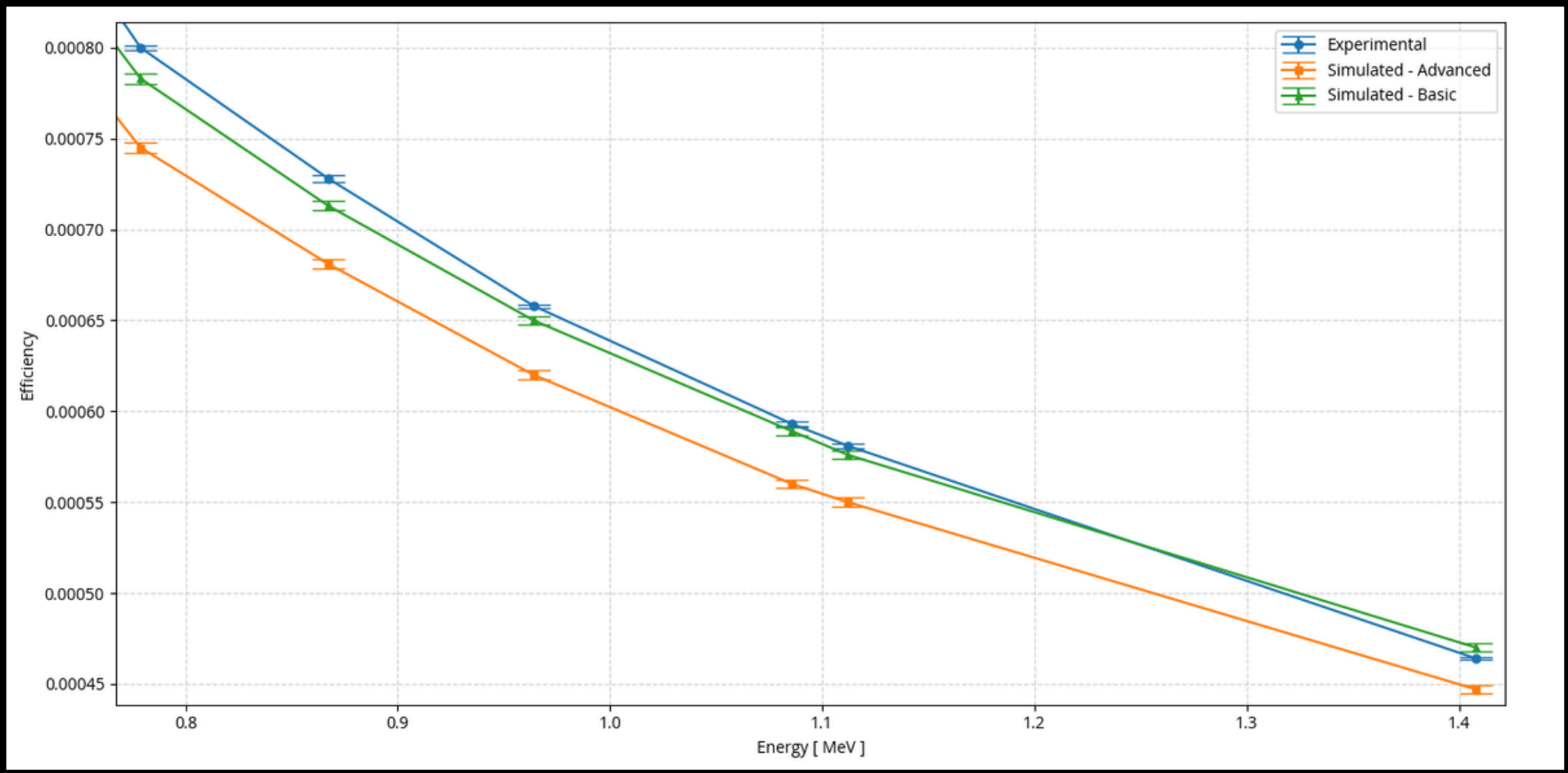
Table 2.- Simulated (advanced model) and experimental full-energy peak efficiencies for ¹⁵² Eu.			
Energy [MeV]	Experimental Efficiency (x10 ⁻³)	Simulated – Advanced Efficiency (x10 ⁻³)	Discrepancy (%)
0.12178	3.660±0.002	2.770±0.017	24.45
0.24469	2.540±0.003	2.160±0.015	15.03
0.34427	1.780±0.001	1.570±0.013	11.98
0.41111	1.500±0.004	1.340±0.004	10.94
0.44398	1.370±0.004	1.240±0.003	9.44
0.77889	0.800±0.001	0.745±0.003	6.83
0.86732	0.728±0.002	0.681±0.003	6.52
0.96401	0.658±0.001	0.620±0.002	5.84
1.08578	0.593±0.001	0.560±0.002	5.51
1.11202	0.581±0.001	0.550±0.002	5.29
1.40795	0.464±0.001	0.447±0.002	3.66

The Advanced Model consistently underestimates experimental efficiency, resulting in higher discrepancies across all energies (average of 9.6%, reaching 24.45% at 0.122 MeV).



RESULTS

High Energies: A Nearly Constant Relative Difference



Detail of the experimental and simulated efficiency curves at high energies (>0.75 MeV).

vs. experiment, above ~750 keV:
< 3% (basic) / < 7% (advanced)

Table 3.- Relative difference between the basic and advanced simulated efficiencies for ¹⁵²Eu, and their respective computational times.

Energy [MeV]	Basic-Advanced Difference (%)	Computational Time – Basic [min]	Computational Time – Advanced [min]
0.12178	6.50	164.60	192.57
0.24469	5.31	81.93	96.72
0.34427	5.57	60.28	78.59
0.41111	4.95	562.43	714.11
0.44398	5.00	548.34	705.34
0.77889	5.05	561.84	801.10
0.86732	4.72	573.68	838.86
0.96401	4.96	584.15	883.43
1.08578	5.11	604.27	932.33
1.11202	4.77	605.09	955.03
1.40795	5.09	636.22	1055.10

The basic-advanced difference remains close to 5% across the investigated energy range, reflecting the combined effect of the modeled geometric differences.



RESULTS

Investigating Discrepancy Sources



Dead Layer (excluded from both models)

Excluded from both models. Low-energy discrepancies are large and strongly energy-dependent, but their direction doesn't match what a missing dead layer alone would predict, so the origin remains an open question.



Crystal & Electrode-Bore Geometry

Produces a small but essentially constant ~5% bias between the two simulated models, across the entire energy range investigated, a systematic offset, not a localized effect.

→ ***Confirming whether these are truly independent, separable effects will require future models that explicitly include the dead layer.***



RESULTS

Investigating Discrepancy Sources

Across nearly the entire energy range, the basic model actually shows a smaller discrepancy against the experimental data than the advanced model.

This does NOT mean the simplified geometry is more accurate.

The electrode dimensions were refined iteratively against the Advanced Model using the experimental efficiency at 1.40795 MeV as reference, and then kept unchanged in both models. That this also reduces the basic model's discrepancy against experiment is a by-product of the calibration procedure, not evidence of superior accuracy.

The real indicator of the crystal-edge effect is the **model-vs-model** difference (Table 3), not the model-vs-experiment discrepancy.



CONCLUSIONS

Conclusions and Practical Implications



The difference between the simplified and advanced geometries is approximately 5% in the simulated full-energy peak efficiency across the investigated energy range.



At high energies, the simplified model provides comparable agreement with the experimental data while reducing computational time by approximately 40%. Concretely: 636 vs. 1055 min at 1.40795 MeV, making the basic model the practical choice for high-energy applications.



The results highlight the importance of accurately representing relevant detector geometry when developing HPGe Monte Carlo models.



LIMITATIONS AND FUTURE WORK



Incorporate the detector's dead layer into both models.



Investigate the sensitivity to the internal electrode geometry using experimentally validated dimensions.



Evaluate whether the approximately 5% difference remains after these refinements.



Isolate individual geometric contributions by varying crystal periphery and electrode-bore geometries independently in separate models.



ACKNOWLEDGMENTS

The authors would like to thank the following institutions for their support during the development of this work:

- the Nuclear Engineering Institute (IEN/CNEN);
- the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq);
- the Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro (FAPERJ);
- the Nuclear Simulation and Artificial Intelligence Laboratory (LaSNIA).





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